

Mental Health & Crisis Signal Mining on Social Media

*Multi-class classification and interpretability analysis
on social media mental-health statements*

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Introduction & Problem Definition

1 in 8

people worldwide live with
a mental health disorder
(WHO, 2022)

The problem

- Majority never receive diagnosis — stigma, cost, and access barriers
- Social media (Reddit) provides self-disclosed mental-health signals
- Need: automated NLP-based screening
- Challenge: 7 overlapping classes with shared emotional vocabulary

Our approach

Build a 7-class NLP pipeline comparing classical ML vs fine-tuned transformer

Stress

Depression

Bipolar

Personality D.

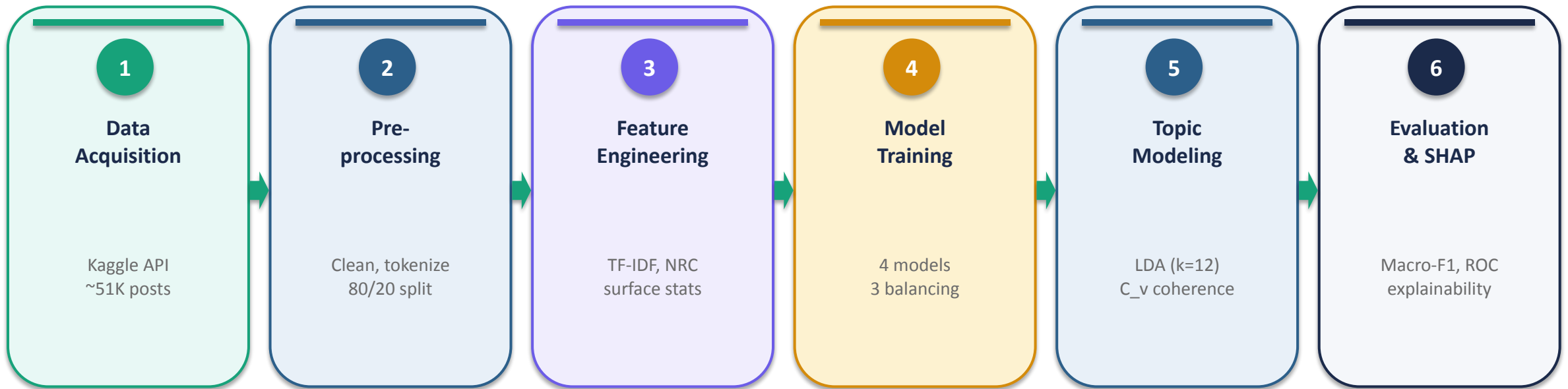
Suicidal

Anxiety

Normal

Models: LogReg • Linear SVM • XGBoost • DistilBERT-LoRA | Interpretability: SHAP + LDA

Method — End-to-End Pipeline



Key design decisions

Primary metric:	Macro-averaged F1 (treats all 7 classes equally)
Data split:	Stratified 80/20, random seed = 42
Leakage check:	Surface-only probe → F1 = 0.22 (no leakage)
Reproducibility:	All notebooks version-controlled on GitHub

Experimental Setup

~52K

statements

7

classes

80/20

train-test

3

balancing

42

seed

EXP-1

Logistic Regression

TF-IDF (uni+bigrams)
3-fold CV, L2
C = 10.0

EXP-2

Linear SVM

TF-IDF (uni+bigrams)
3-fold CV
C = 1.0

EXP-3

XGBoost

TF-IDF + LIWC +
surface + LDA topics
Bayesian search

EXP-4

DistilBERT + LoRA

Raw tokens, 3 epochs
AdamW (lr=2e-5)
Early stopping

Each classical model tested under 3 conditions: no balancing, class-weighted, SMOTE on training split

Results

Final model ranking (best condition per family)

WINNER

DistilBERT + LoRA

Macro-F1 = 0.7139 | Weighted-F1 = 0.7617

#	Model	Condition	M-F1	W-F1	Accuracy	ROC-AUC
1	DistilBERT-LoRA	finetuned	0.7139	0.7617	0.7578	0.9608
2	XGBoost	tfidf+eng+lda	0.7033	0.7500	0.7460	0.9582
3	Logistic Reg.	no_balancing	0.6740	0.7242	0.7223	0.7223
4	Linear SVM	no_balancing	0.6605	0.7141	0.7129	0.7129

Per-class F1: DistilBERT vs XGBoost

Class	BERT	XGB	Winner
Anxiety	0.787	0.775	BERT
Bipolar	0.759	0.762	XGB
Depression	0.693	0.677	BERT
Normal	0.900	0.892	BERT
Pers. Disorder	0.550	0.521	BERT
Stress	0.610	0.615	XGB
Suicidal	0.698	0.681	BERT

DistilBERT wins 5/7 classes

Statistical validation — McNemar's test

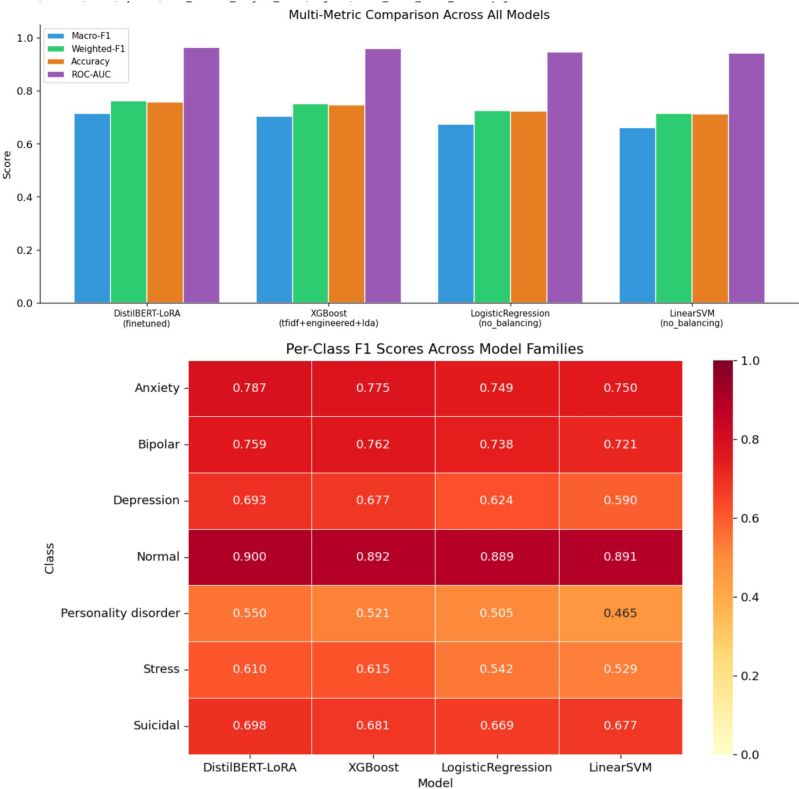
Chi-squared: 8.7451

p-value: 0.003 (significant, $p < 0.05$)

Agreement: 83.6% of test samples

Discordant: 1,730 pairs (16.4%)

The two models have genuinely different error patterns. DistilBERT makes fewer errors on the disagreement set.



Explainability and interpretability

SHAP Attributions

Token-level explanations for individual predictions

- TreeExplainer for XGBoost
- Partition explainer for DistilBERT
- Identifies which words drive each classification decision
- Global feature importance + per-instance force plots

LDA Topic Discovery

Latent topical structure across mental-health classes

- k=12 topics (C_v coherence)
- Topic-by-class cross-tabulation
- Isolation, hopelessness, manic energy, identity disturbance
- pyLDavis interactive HTML visualization

Conclusions & Next Steps

Key findings

Contextual > Bag-of-words

DistilBERT-LoRA outperforms all classical models (+1.1 to +5.3 Macro-F1 points)

XGBoost: strong fallback

Near-BERT performance, <10ms latency, full SHAP interpretability

No leakage detected

Surface-only probe F1 = 0.22 — models learn from content, not formatting

Statistically significant

McNemar's test $p = 0.003$ confirms DistilBERT's superiority is real

Future work

Cross-platform generalization — test on additional mental-health sources

Domain-specific transformers — MentalBERT, mental-RoBERTa

Longitudinal modeling — temporal user trajectories with RNNs

Clinical validation — collaborate with professionals vs DSM-5

Active learning — feedback loop to reduce labeling cost